



EMISSIONS REPORT

MARCH 2025 EMISSIONS REVIEW –
COMPUTING FOCUS

---- For Executive Usage Only ----

MERIDAN TRUST BANK

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INTRODUCTION

The purpose of this report is to understand, review, and categorize the March 2025 carbon emissions of Meridian Trust Bank (MTB).

Page 2 of this document outlines the total carbon emissions relevant to MTB's operations using the Greenhouse Gas Protocol (GHG). Carbon emissions reflect a monthly total and are itemized by source and scope. Scope 1 emissions consist of direct emissions from company-owned sources, such as vehicles. Scope 2 emissions are those from purchased energy sources such as electricity. Scope 3 emissions consist of all other indirect emission sources not already covered in Scopes 1 or 2.

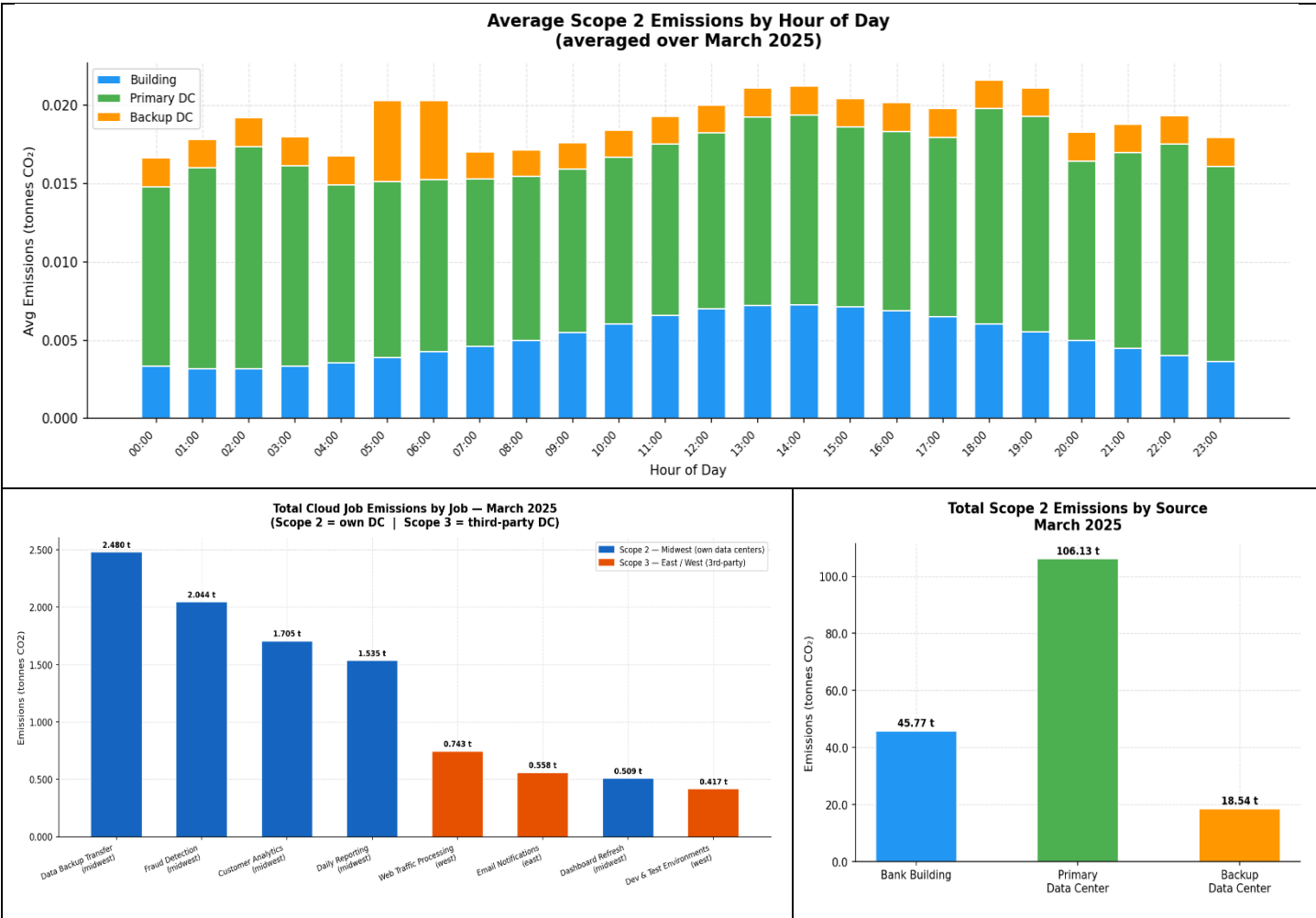
Page 3 of this document focuses on the impact of carbon emissions of MTB's ICT sector and scores MTB using the Green Software Foundation's Green Software Maturity Matrix (GSMM). The GSMM is a widely used tool designed to rate organizations on their progress towards fully green practices and to provide guidance on the best next steps.

CARBON ASSESSMENT

The table below itemizes carbon emissions by source category and scope. Carbon emissions are expressed in tonnes of CO₂ released throughout the month of March 2025. Over the month, all sectors of MTB emitted approximately **183.7 tonnes** of CO₂. On-site scope 2 emissions due to electricity usage constitute the majority of MTB’s carbon emissions. The top chart details the average scope 2 emissions by hour in March 2025. The bottom right chart presents the total carbon emissions of MTB’s properties over March 2025. The bottom left chart shows the total emissions from various daily computing tasks performed by MTB ICT.

Source	Scope 1 (tonnes CO ₂)	Scope 2 (tonnes CO ₂)	Scope 3 (tonnes CO ₂)
Fuel	11.6	-	-
Bank Buildings	-	45.8	-
Primary Data Center	-	106.1	-
Secondary Data Center	-	18.5	-
Cloud Jobs*	-	8.3*	1.7
Total – 183.7 tonnes CO ₂	11.6	170.4	1.7

* Scope 2 cloud jobs overlap with other categories and are not double counted in the total



COMPUTING-FOCUSED FOOTPRINT

This section of the report outlines the current carbon emissions of MTB attributable to ICT operations and scores MTB on the Green Software Maturity Matrix. Recommendations to improve MTB's GSMM level, and to reduce total carbon emissions, will be provided in subsequent documents.

ICT-FOCUSED EMISSIONS

Emissions from the Information and Communication Technology (ICT) sector currently compose approximately **68.7%** of MTB's emissions. Of this percentage, approximately **84%** of ICT emissions can be attributed to the electricity used to power MTB's primary data center and ensure high uptime. The 68.7% of emissions due to ICT at MTB is far greater than the standard 1.5-4% that can be attributed to ICT globally.

GREEN SOFTWARE MATURITY MATRIX (GSMM)

The table below details MTB's current state on the industry-standard GSMM. Information on the GSMM is detailed in the introduction section of the document. MTB is currently positioned at **Level 1**, "Aspiring", on the GSMM. Due to the relative novelty of the GSMM in the banking industry, Level 1 is currently standard for most other banks. MTB currently performs best, at Level 3, in the carbon-metrics and monitoring section of the GSMM due to the 5-minute energy usage updates available for data centers and banks.

With the creation of this document outlining scope 1 and scope 2 emissions, MTB scores at Level 2 of the carbon footprint section of the GSMM. Finally, due to MTB's continued usage of legacy devices in data centers, thus reducing embodied carbon costs, MTB is currently at Level 2 of the device section of the GSMM. MTB is currently at Level 1 of all other sections of the GSMM.

	ASPIRING	AWARE	ACTING	AWESOME	INSPIRING
 commitments	none	carbon neutral	carbon zero with offsets	10% (offset)	1% (offset)
 footprint	unknown	know scope 1&2	reducing per unit	reducing absolutely	~zero
 metrics	none	report scope 1&2	daily scope 1&2&3	realtime	predicted
 carbon ops	none	manual	lightswitch ops	auto-rightsizing	carbon SRE
 energy	none	green hosting	dynamic management	demand shaping	24/7 Carbon Free Electricity
devices	none	some targets	10y/90%	10y/100%	rolling repair
 utilization	none	some multi-tenant	all multi-tenant	max orchestration	edge integration
products	none	carbon awareness	demand shaping	feature tracking	feature carbon error budgets
 training	ad hoc	basic/champions	advanced	you are the trainer	you are the leader